

Proposed Research Topic

Single-Agent vs. Multi-Agent LLM Systems: Evaluating Agentic Architectures for Reasoning Performance and Computational Efficiency

Large Language Models (LLMs) are increasingly being used in agentic systems where multiple AI agents collaborate, debate, delegate tasks, or review each other's responses. Although multi-agent architectures are often claimed to improve reasoning and reliability, adding more agents also increases token usage, execution time, communication overhead, and computational cost. This research will investigate whether multi-agent LLM systems actually provide meaningful reasoning improvements over single-agent systems, particularly when computational resources are taken into account. The main research question will be: Under comparable computational budgets, when do multi-agent architectures outperform a single-agent LLM, and when does adding additional agents produce diminishing or negative returns?

The project will use Google Agent Development Kit (ADK) to implement several controlled agentic architectures using the same underlying LLM where possible. These will include a baseline single-agent system and multiple multi-agent structures, such as parallel independent agents with voting, sequential agents where one agent reviews or improves another's answer, and debate-based architectures where agents critique competing solutions before a final response is selected. The different architectures will be evaluated against established reasoning benchmarks such as GSM8K, MMLU-Pro, BIG-Bench Hard (BBH), or similar datasets. Performance will be measured using reasoning accuracy, token consumption, latency, computational/API cost, consistency, and improvement relative to the number of agents used. Experiments will also examine whether multi-agent benefits vary according to task difficulty. The expected outcome is an empirical comparison showing which agentic structures provide genuine reasoning improvements and whether those improvements justify their additional computational cost. Rather than simply building a multi-agent application, the research will use Google ADK as an experimental framework to systematically compare agent architectures and investigate the relationship between agent collaboration, reasoning performance, and computational efficiency.